ADDITIONAL QUESTIONS

LIMITS, CONTINUITY, DIFFERENTIABILITY

SINGLE CORRECT ANSWER TYPE

1. **Let** $f(x)=\lim_{n\to\infty}\sum_{r=1}^{n}3^{r-1}\sin^3\frac{x}{3^r}$ **and**

g(x)=x-4f(x) then $\lim_{x\to 0}(1+g(x))^{\cot x}$ **is**

A) $\frac{1}{e}$ B) e^2 C) e D) 1

2. **Let** $f(x)$ **be a real valued function defined on the interval** $(0,\infty)$ **by**

$f(x)=\ln x+\int_0^x\sqrt{1+\sin t}\,dt$. **Then which of the following statements is (are) true**

[IIT-2010]

(A) $f^{11}(x)$ exists for all $x\in(0,\infty)$

(B) $f^1(x)$ exists for all $x\in(0,\infty)$ and $f^1(x)$ is continuous on $(0,\infty)$ but not differentiable on $(0,\infty)$

(C) There exists $\alpha>1$ such that $\left|f^1(x)\right|<\left|f(x)\right|$ for all $x\in(\alpha,\infty)$

(D) There exists $\beta>0$ such that $\left|f(x)\right|+\left|f^1(x)\right|\le\beta$

3. **Let f(x) = degree of** $\left(u^{x^2}+u^2+2u-3\right)$ **(where u is a real parameter, x is variable) then**

A) f(x) is continuous at $x=\sqrt{2}$ but not differentiable

B) f(x) is neither differentiable nor continuous at $x=\sqrt{2}$

C) f(x) is discontinuous at $x=\sqrt{2}$

D) f(x) is differentiable at $x=\sqrt{2}$

4. **Given**

$$f(x)=\left\{\log_a\left(a\left|[x]+[-x]\right|\right)^x\left(\frac{a^{\left(\frac{[x]+[-x]}{|x|}\right)^{\frac{2}{-5}}}}{3+a^{\frac{1}{|x|}}}\right);\right.$$

$|x|\neq 0, a>1$ **and f(x)=0 for x = 0 (where [.] is G..I.F) then**

A) f is continuous but not differentiable at $x=0$

B) f is continuous and differentiable at $x=0$

C) The differentiability of f at $x=0$ depends on the value of a

D) f is continuous and differentiable at $x=0$ and for $a=e$ only

5. **Let f(0) = 0 and** $f^1(0)=1$ **. For a positive integer K.** $\lim_{x\to 0}\frac{1}{x}\left(f(x)+f\left(\frac{x}{2}\right)+.....+f\left(\frac{x}{k}\right)\right)$ **is**

A) $\frac{k(k+1)}{2}$ B) $\frac{k+1}{2k^2}$

C) $1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}$ D) $\frac{1}{2}$

6. **Let** $f:[a,b]\to[1,\infty)$ **be a continuous function and** $g:R\to R$ **be defined as**

$$g(x)=\begin{cases}0, & if \quad x<a\\ \int_a^x f(t)\,dt, if & a\le x\le b. Then,\\ \int_a^b f(t)\,dt, if & x>b\end{cases}$$

(A) g(x) is continuous but not differentiable at a

(B) g(x) is differentiable on R

(C) g(x) is continuous but not differentiable at b

(D) g(x) is continuous and differentiable at either a or b but not both

MULTI ANSWER QUESTIONS

7. **Which of the following statement(s) is/are TRUE?**

Statement A: If function $y=f(x)$ is continuous at x = c such that $f(c)\neq 0$ then $f(x)f(c)>0\,\forall x\in(c-h,c+h)$ where h is sufficiently small positive quantity.

Statement B:

$$\lim_{n\to\infty}\frac{1}{n}\ln\left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)......\left(1+\frac{n}{n}\right)\right]=1+2\ln 2$$

Statement C: Let f be a continuous and non-negative function defined on [a, b]. if $\int_a^b f(x)dx=0$ then $f(x)=0\ \forall\ x\in[a,b]$.

Statement D: Let f be a continuous function defined on [a, b] such that $\int_a^b f(x)dx=0$, then there exists atleast one $c\in(a,b)$ for which $f(c)=0$.

8. **If $a,b,c\in R^{+}$ then $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{(k+an)(k+bn)}$ is equal to**

A) $\frac{1}{a-b}\ln\frac{b(b+1)}{a(a+1)}$ if $a\neq b$

B) $\frac{1}{a-b}\ln\frac{a(b+1)}{b(a+1)}$ if $a\neq b$

C) non existent if $a=b$

D) equals $\frac{1}{a(1+a)}$ if $a=b$

9. **Let f(x) =** $\begin{cases}\int_0^x(1+|1-t|)dt & , x>2\\ 5x+1 & , x\leq 2\end{cases}$ **, then**

(A) f(x) is not continuous at $x=2$

(B) f(x) is continuous but not differentiable at $x=2$

(C) f(x) is differentiable everywhere

(D) the right derivative of f(x) at $x=2$ does not exist

10. **If $f'(x)=f(x)+\int_{-10}^{10}f(x)dx$, then**

(A) $f'(x)=f(x)$ if $\lim_{x\to\infty}f(x)=0$

(B) $f'(x)-f(x)=c$ (any constant)

(C) $f''(x)-f(x)=c$ (any constant)

(D) None of the above

11. **Let f(x) be a non–constant twice differentiable function defined on (–∞, ∞) such that f(x) = f(1–x) and $f'\left(\frac{1}{4}\right)=0$.**

Then, [IIT-2008]

(A) f″(x) vanishes at least twice on [0, 1]

(B) $f'\left(\frac{1}{2}\right)=0$

(C) $\int_{-1/2}^{1/2}f\left(x+\frac{1}{2}\right)\sin x\,dx=0$

(D) $\int_0^{1/2}f(t)e^{\sin\pi t}dt=\int_{1/2}^{1}f(1-t)\,e^{\sin\pi t}\,dt$

COMPREHENSION TYPE

Passage : L' Hospital's rule has many versions one them is this suppose $f,g:(a,b)\to R$ are differentiable on (a, b). Suppose further that

i) $g^1(x)\neq 0$ **for** $x\in(a,b)$

(ii) $\lim_{x\to a^+}g(x)\to\infty(\text{or}-\infty)$

(iii) $\lim_{x\to a^+}\frac{f^1(x)}{g^1(x)}=L$ **then** $\lim_{x\to a^+}\frac{f(x)}{g(x)}=L$

(This rule can be extended to cover the case where a and b tends to infinity or L tends to infinity)

12. **Let 'f' be a differentiable function on** $(0,\infty)$.

If $\lim\limits_{x\to\infty}\left(\sin\left(\frac{\pi}{10}\right)f(x)+f^1(x)\right)=\sec\left(\frac{\pi}{5}\right)$ **then** $\lim\limits_{x\to\infty} f(x)$ **equals**

A) $\frac{1}{4}$ B) 4 C) $3-\sqrt{5}$ D) $\frac{3+\sqrt{5}}{4}$

13. **Let 'f' be a differentiable function on** $(0,\infty)$.

If $\lim\limits_{x\to\infty}\left(\tan\left(\frac{\pi}{8}\right)f(x)+2\sqrt{x}f^1(x)\right)=\cot\frac{\pi}{12}$ **then** $\lim\limits_{x\to\infty} f(x)$ **equals.**

A) $\sqrt{8}-\sqrt{6}+\sqrt{4}-\sqrt{3}$

B) $\sqrt{8}+\sqrt{6}-\sqrt{4}-\sqrt{3}$

C) $\sqrt{3}+\sqrt{4}+\sqrt{6}+\sqrt{8}$

D) $\sqrt{8}-\sqrt{6}-\sqrt{4}+\sqrt{3}$

Passage : Let 'f' be a polynomial function satisfying

f(x).f(y)+2=f(x)+f(y)+f(xy) $\forall x,y\in R^+\cup\{0\}$ **and**

f(x) is one-one with f(0)=1 and $f^1(1)=2$

then answer the following

14. **The function y = f(x) is given by**

A) x^2-1 B) $x^{1/3}-1$

C) $1+\frac{2x^3}{3}$ D) $1+x^2$

15. **The number of points of non-differentiability of** $h(x)=\min\left\{\frac{2}{f(x)},x^2,\left|1-|x|\right|\right\}$, $x\geq 0$ **is**

A) 6 B) 5 C) 3 D) 4

MATCHING TYPE QUESTIONS

16. **Match the following :**

Column I

A)

$$f(x)\begin{cases}\dfrac{\left(\frac{\pi}{2}-\sin^{-1}\left(1-\{x\}^2\right)\right)\left(\sin^{-1}\left(1-\{x\}\right)\right)}{\sqrt{2}\left(\{x\}-\{x\}^3\right)} & x>0\\ k & x=0\\ \dfrac{A\sin^{-1}\left(1-\{x\}\right)\cos^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}\left(1-\{x\}\right)}} & x<0\end{cases}$$

is continuous at $x=0$ then the value of

$\sin^2 k+\cos^2\left(\frac{A\pi}{\sqrt{2}}\right)$ is

B) $f(x)=\left[2+5|n|\sin x\right]n\in Z$ has exactly 19 points of non differentiability in $x\in(0,\pi)$ then possible values of n are

C) If $f(x)=\begin{cases}x.\dfrac{(3/4)^{1/x}-(3/4)^{-1/x}}{(3/4)^{1/x}-(3/4)^{-1/x}} & x\neq 0\\ 0 & x=0\end{cases}$

if $P=f'(0^-)-f'(0^+)$ then

$\lim\limits_{x\to P^-}\dfrac{\left(\exp(x+2)\ln 4\right)^{\frac{[x+1]}{4}}-16}{4^x-16}$ is $\leq$

Column II

P) 2

Q) –2

R) 3

S) –3

INTEGER TYPE

17. Number of points where the function f(x) = $\left(x+\left[x^3+1\right]\right)^{x^2+\sin x}$ is not differentiable for $x\in(1,3/2)$ is (where [.] is G.I.F)

18. Let $f:R^+\to R$ satisfies the functional equation $f(xy)=e^{xy-x-y}\left(e^y f(x)+e^x f(y)\right)$ $\forall x,y\in R^+$ and $f^1(1)=e$ then $\left[\ln(f(e))\right]$ (where [.] is G.I.F) is

19. Let $I_n=\int_{-1}^{1}|x|\left(1+x+\frac{x^2}{2}+\frac{x^3}{3}+....+\frac{x^{2n}}{2n}\right)dx$.

If $\lim\limits_{n\to\infty} I_n$ can be expressed as rational $\frac{p}{q}$ in the lowest form, then the value of pq is

SUBJECTIVE QUESTIONS

20. Let f (x) = $x^3 - x^2 + x + 1$ and

$$g(x)=\begin{cases}\max\{f(t);\ 0\le t\le x\}, & 0\le x\le 1\\ 3-x; & 1<x\le 2\end{cases}$$

Discuss the continuity and differentiability of the function g(x) in the interval (0, 2).

21. Show that the transformation z = log tan (x/2) reduces the differential equation

$$\frac{d^2y}{dx^2}+\cot x\frac{dy}{dx}+4y\,\mathrm{cosec}^2x=0 \text{ into } \frac{d^2y}{dz^2}+4y=0.$$

22. Let $f(x)=\begin{cases}1+x & 0\le x\le 2\\ 3-x & 2<x\le 3\end{cases}$

Determined g(x) = f (f(x)) and hence find the points of discontinuity of g, if any.

23. Let f(x) be defined in the interval [-2, 2] such that $f(x)=\begin{cases}-1, & -2\le x\le 0\\ x-1, & 0<x\le 2\end{cases}$ and $g(x)=f(|x|)+|f(x)|$. Test the differentiability of g(x) in (-2, 2).

24. Let R be the set of real numbers and $f:R\to R$ be such that for all x and y in R, $|f(x)-f(y)|\le|x-y|^3$. Prove that f(x) is a constant.

25. Draw a graph of the function, $y=[x]+|1-x|$, $-1\le x\le 3$. Determine the points, if any, where this function is not differentiable, where [.] denotes the greatest integer function.

26. Let f(xy) = f(x) f(y) for all x ,y $\in$ R. If the function f(x) is continuous at x = 1, show that f(x) is continuous for all x $\neq$ 0.

27. If $f\left(\frac{x+y}{3}\right)=\frac{2+f(x)+f(y)}{3}$ for all real x and y and $f'(0)=2$, then determine y = f(x).

28. Let $f:R\to R$ and $g:R\to R$ be respectively given by $f(x)=|x|+1$ and $g(x)=x^2+1$. Define $h:R\to R$ by

$$h(x)=\begin{cases}\max\{f(x),g(x)\}, & if\ x\le 0.\\ \min\{f(x),g(x)\}, & if\ x>0\end{cases}$$

The number of points at which $h(x)$ is not differentiable is

29. Let the variable x_n be determined by the following law of formation $x_0=\sqrt{a}$

$x_1=\sqrt{a+\sqrt{a}}$, $x_2=\sqrt{a+\sqrt{a+\sqrt{a}}}$

$x_3=\sqrt{a+\sqrt{a+\sqrt{a+\sqrt{a}}}}$, then find $\lim\limits_{n\to\infty} x_n$.

30. A function $f:R\to R$ satisfies the equation f(x + y) = f(x) . f(y) for all x, y in R and $f(x)\neq 0$ for any x in R. Let the function be differentiable at x = 0 and $f'(0)=2$. Show that $f'(x)=2f(x)$ for all x in R. Hence determine f(x). [IIT-1990]

31. If f(x) = $\sqrt{|x-1|}$ and g(x) = sin x then calculate (fog) (x) and (gof) (x) and discuss the differentiability of (gof) (x) at x = 1. [IIT- 92]

32. Let f [(x + y) / 2] = {f(x) + f(y)} / 2 for all real x and y. If $f'(0)$ exists and equals – 1 and

f(0) = 1, find f(2). [IIT - 95]

33. (i) $\lim\limits_{x\to 0}\frac{2(\tan x-\sin x)-x^3}{x^5}$.

(ii) $\lim\limits_{x\to 1}\frac{(\ell n(1+x)-\ell n2)(3.4^{x-1}-3x)}{\left[(7+x)^{\frac{1}{3}}-(1+3x)^{\frac{1}{2}}\right].\sin(x-1)}$

[IIT - 97]

(iii) Through a point A on a circle, a chord AP is drawn and on the tangent at A a point T is taken such that AT = AP. If TP produced meet the diameter through A at Q then prove that the limiting value of AQ when P moves upto A is double the diameter of the circle.

34. Let $\alpha\in R$. Prove that a function $f:R\to R$ is differentiable at α if and only if there is a function $g:R\to R$ which is continuous at α and satisfies f(x) - f(α) = g(x) (x-α) for all $x\in R$. [IIT - 2001]

35. If a function $f:[-2a,\,2a]\to R$ is an odd function such that f(x) = f(2a - x) for $x\in[a,\,2a]$ and the L H D at x = a is zero then find the L H D at x = -a. [IIT - 2003]

36. $f'(0)$ can be represented for any differentiable function $f(-1,\,1)\to R$ as

$f'(0)=\lim\limits_{x\to\infty} nf\left(\frac{1}{n}\right)$, where f(0) = 0. Find

$\lim\limits_{n\to\infty}\left[\frac{n+1}{\pi/2}\cos^{-1}\left(\frac{1}{n}\right)-n\right]$,given

$\cos^{-1}\left(\frac{1}{n}\right)\le\frac{\pi}{2}\ \forall\ n\in N.$

37. If f(x – y) = f(x) . g(y) – f(y) . g(x) and g(x – y) = g(x) . g(y) + f(x) . f(y) for all $x,\ y\in R$. If right hand derivative at x = 0 exists for f(x), then find the derivative of g(x) at x = 0.

38. Show that the function $f:R\to R$ defined by $f(x)=x\left\{1+\frac{1}{3}\sin\left(\ln x^2\right)\right\};x\ne 0$ and $f(0)=0$ is everywhere continuous, but has no differential coefficient at the origin.

39. Let f be a function satisfying

$f(x+y)+\sqrt{6-f(y)}=f(x)f(y)$ and

$f(h)\to 6$ as $h\to 0$.

Discuss the continuity of f.

40. Let f be a function such that

$f(x+f(y))=f(f(x))+f(y)\quad\forall x,y\in R$

and $f(h)=h\ \ for\ 0<h<\varepsilon\ where\ \varepsilon>0,$

then determine $f^{/}(x)$ and $f(x)$.

41. The value of

$\lim\limits_{n\to\infty}\left(\frac{1}{2n+1}+\frac{1}{2n+3}+\frac{1}{2n+5}+.......+\frac{1}{4n-1}\right)$ is

$\frac{a}{b}\ln c$ where $a,b,c\in N$. Then the least value of $a+b+c$ is

KEY

1) C 2) B 3) A 4) B
5) C 6) D 7) A,C,D 8) B,D
9) A,D 10) A,B,C 11) A,B,C,D
12) B 13) C 14) D 15) C
16) A-P, B- P, Q, C-Q, R
17) 2 18) 2 19) 6

22. Continuous in (0, 2) and differentiable in (0, 2) except at x = 1

23. $g(x)=\begin{cases}2+x & 0\le x\le 1\\ 2-x & 1<x\le 2\\ 4-x & 2<x\le 3\end{cases}$

discontinuity at x = 1, 2 & non differentiable at x = 0, 1

25. {0, 1,2,3}

27. 2x + 2

28. $g(x)$ is continuous for all $x\in[0,2]$

but $g(x)$ is not differentiable at x = 1

29. 3

30. e^{2x}

31. not differentiable at x=1

32. –1

33. (i) 1/4 , (ii) $-\frac{9}{4}\ell n\frac{4}{e}$

34. f(x) is differentiable at $x=\alpha$
If g(x) is continuous at $x=\alpha$.

35. 0

36. $\frac{\pi-2}{\pi}$

37. 0

39. If $f(x)\neq 0$, then f is discontinuous at x. If $f(x)=0$, then $f(x)$ is continuous at x.

40. $f(x)=x$ & $f'(x)=1$

41. 5

HINTS

1. $f(x)=\frac{1}{4}\underset{n\to\infty}{\text{Lt}}\sum_{r=1}^{n}\left(3^r\sin\frac{x}{3^r}-3^{r-1}.\sin\frac{x}{3^{r-1}}\right)$

$=\frac{1}{4}\underset{n\to\infty}{\text{Lt}}\left(3\sin\frac{x}{3}-\sin x+3^2\sin\frac{x}{3^2}\right.$

$\left.-3\sin\frac{x}{3}+...+3^n\sin\frac{x}{3^n}-3^{n-1}\sin\frac{x}{3^{n-1}}\right)$

$=\frac{1}{4}\underset{n\to\infty}{\text{Lt}}\left(3^n\sin\frac{x}{3^n}-\sin x\right)$

$=\frac{1}{4}\left(\underset{n\to\infty}{\text{Lt}}\frac{\left(\sin\frac{x}{3^n}\right)}{\left(\frac{x}{3^n}\right)}.x-\sin x\right)=\frac{x-\sin x}{4}$

$g(x)=x-4+\sin x=\sin x$

$\underset{x\to 0}{\text{Lim}}(1+\sin x)^{\cot x}=e^{\sin x.\frac{\cos x}{\sin x}}=e$

2. $f^1(x)=\frac{1}{x}+\sqrt{1+\sin x}$ exists and continuous $\forall x\in(0,\infty)$

$f^{11}(x)=-\frac{1}{x^2}+\frac{\cos x}{2\sqrt{1+\sin x}}$

$f^1(x)=$ is not differentiable at $\sin x=-1$

where $x=2n\pi-\frac{\pi}{2}$

$f^1(x)$ is not differentiable at $\sin x=-1$

where $x=2n\pi-\frac{\pi}{2}$

$f(x)$ and $f^1(x)$ are positive for all $x\in(\pi,\infty)$

$f(x)>f^1(x)\Rightarrow\}|f(x)|>|f^1(x)|$

$\frac{1}{x}$ is not bounded hence D is not correct

3. f(x) = 2 if $x\le\sqrt{2}$

$=x^2$ if $x>\sqrt{2}$

$\therefore\underset{x\to\sqrt{2}^+}{\text{Lt}}f(x)=\underset{x\to\sqrt{2}^+}{\text{Lt}}x^2=2,\ \underset{x\to\sqrt{2}^-}{\text{Lt}}f(x)=2\Rightarrow$ It is continuous at $x=\sqrt{2}$

$f^1\left(\sqrt{2}^+\right)=\underset{h\to 0^+}{\text{Lt}}\frac{f\left(h+\sqrt{2}\right)-f\left(\sqrt{2}\right)}{h}$

$=\underset{h\to 0^+}{\text{Lt}}\frac{\left(h-\sqrt{2}\right)^2-2}{h}=2\sqrt{2}$

$f^1\left(\sqrt{2}^-\right)=\underset{h\to 0^-}{\text{Lt}}\frac{f\left(h+\sqrt{2}\right)-f\left(\sqrt{2}\right)}{h}$

$=\underset{h\to 0^-}{\text{Lt}}\frac{2-2}{h}=0$

4. $f^1\left(\overset{+}{0}\right)=\underset{h\to 0^+}{\text{Lt}}\left(h.\log_a a|-1|\right)^h\left(\frac{a^{\frac{\frac{2}{0-1}-5}{h}}}{3+a^{1/h}}\right)=0$

similarly $f^1(0^-)=0$.

$\therefore$ It is differentiable and continuous at $x=0$

5. $\underset{x\to 0}{\text{Lt}}\frac{1}{x}\left(f(x)+f\left(\frac{x}{2}\right)+.....+f\left(\frac{x}{k}\right)\right)$

$=\underset{x\to 0}{\text{Lim}}\left(\frac{f(x)}{x}+\frac{f\left(\frac{x}{2}\right)}{x}+....+\frac{f\left(\frac{x}{k}\right)}{x}\right)$

$=f^1(0)+\frac{1}{2}f^1(0)+.....+\frac{1}{k}f^1(0)$

$$=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}$$

6. (d)Continuity of g(x) at x=a

$$\lim_{x\to a^-} g(x)=0 \ and \ \lim_{x\to a^+}\int_a^x f(t)dt=\int_a^a f(t)dt=0$$

Also, $g(a)=\int_a^a f(t)dt=0$

Similarly, at x=b we can prove that g(x) is continuous.

Thus, g(x) is continuous, $\forall x\in R$.

Now, $g'(x)=\begin{cases}0 & , \ x<a\\ f(x), & a\le x\le b\\ 0 & , \ x>b\end{cases}$

$$g'(a^-)=0, g'(b^-)=f(b)$$

$$g'(a^+)=f(a), g'(b^+)=0$$

f(a) and f(b) can never be zero as codomain of f(x) is $[1,\infty)$.

Hence, g(x) is non-derivable at x=a and x=b

7. $$f(x)=\sum_{r=1}^{n}\left(x^{2r}+\frac{1}{x^{2r}}+2\right)=\sum_{r=1}^{n}x^{2r}+\sum_{r=1}^{n}\frac{1}{x^{2r}}+2n$$

$$=\left(x^2+x^4+......+x^{2n}\right)+\left(\frac{1}{x^2}+\frac{1}{x^4}+.....+\frac{1}{x^{2n}}\right)+2n$$

$$=\frac{x^2\left(1-x^{2n}\right)}{\left(1-x^2\right)}+\frac{1}{x^2}\frac{\left(1-\frac{1}{x^{2n}}\right)}{1-\frac{1}{x^2}}+2n$$

$$f(x)=\frac{x^2\left(1-x^{2n}\right)}{1-x^2}+\frac{\left(1-x^{2n}\right)}{\left(1-x^2\right)x^{2n}}+2n$$

$$f(x)=\frac{\left(1-x^{2n}\right)}{1-x^2}\left(x^2+\frac{1}{x^{2n}}\right)+2n \quad x\ne\pm1$$

$$\therefore \ \left(f(x)-2n\right)\left(1-x^2\right)=\left(1-x^{2n}\right)\left(x^2+\frac{1}{x^{2n}}\right)$$

now consider

$g(x)=\lim_{n\to\infty}\left(\left(f(x)-2n\right)x^{-2n-2}\left(1-x^2\right)\right)$ for $x\ne\pm1$

$$=\lim_{n\to\infty}\left(1-x^{2n}\right)\left(x^2+\frac{1}{x^{2n}}\right).\frac{1}{x^{2n+2}};\ x\ne\pm1$$

$$=\lim_{n\to\infty}\frac{\left(1-x^{2n}\right)\left(x^{2n+2}+1\right)}{\left(x^{2n}\right)\left(x^{2n+2}\right)}$$

$$=\lim_{n\to\infty}\left(-1+\frac{1}{x^{2n}}\right)\left(1+\frac{1}{x^{2n+2}}\right)$$

now $g(x)=\begin{bmatrix}-1 & if \ |x|>1\\ -\infty & if \ |x|<1\\ 0 & if \ x=\pm1\end{bmatrix}$

now clearly g has non removable infinite type of discontinuity at $x=1$ and -1

g is continuous at $x=2$

8. $$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{n}{n^2\left(a+\frac{k}{n}\right)\left(b+\frac{k}{n}\right)}$$

$$=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\frac{1}{\left(a+\frac{k}{n}\right)\left(b+\frac{k}{n}\right)}$$

$$=\int_0^1\frac{1}{(a+x)(b+x)}dx=\frac{1}{a-b}\int_0^1\frac{(a+x)-(b+x)}{(a+x)(b+x)}dx$$

$$=\frac{1}{a-b}\int_0^1\left(\frac{1}{b+x}-\frac{1}{a+x}\right)dx$$

$$=\frac{1}{a-b}\left[\ln(b+x)-\ln(a+x)\right]_0^1=\frac{1}{a-b}\left[\ln\frac{(b+x)}{(a+x)}\right]_0^1$$

$$=\frac{1}{a-b}\ln\frac{(b+1)a}{(a+1)b}$$

if $a=b$,

$$l=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{\left(a+\frac{k}{n}\right)^2}=\int_0^1\frac{dx}{\left(a+x\right)^2}=\left.\frac{1}{a+x}\right]_1^0$$

$$=\frac{1}{a}-\frac{1}{a+1}=\frac{1}{a\left(a+1\right)}$$

9. $f\left(x\right)=\begin{cases}5x+1, & x\le 2\\ \frac{x^2}{2}+1, & x>2\end{cases}$, so f(x) is not continuous at $x=2$.

10. key: ABC

$f''\left(x\right)=f'\left(x\right)$

$f'\left(x\right)=C_1e^x$

$f\left(x\right)=C_1e^x+C_2$

11. (A) $f'(x)=-f'(1-x)$

$f'(1/4)=-f'(3/4)=0$

$f'(3/4)=0$

$f'(1/4)=0$

$\Rightarrow \exists$ a point $\in$ (1/4, 3/4) in which $f''(x)=0$

$f''(1/2)=0$

(B) $f(x)=f(1-x)$

$f'(x)=-f'(1-x)$

$f'(1/2)=-f'(1/2)$

$f'(1/2)=0.$

(C) $f(x+1/2)=f(1-x-1/2)\Rightarrow f(x+1/2)=f(1/2-x)$

i.e., $f(x+1/2)$ is even

i.e., $\int_{-1/2}^{1/2} f(x+1/2)\sin x=0$

(D) $\int_0^{1/2} f(t)\, e^{\sin\pi t}dt=\int_{1/2}^{1} f(1-t)\ e^{\sin\pi t}\ dt$

Put $(1-t)=u$ in RHS

$\Rightarrow$ R.H.S. $=-\int_{1/2}^{0} f(u)\ e^{\sin\pi(1-u)}\ du\ =\int_0^{1/2} f(u)\ e^{\sin\pi u}\ du$

12. $\lim_{x\to\infty} f\left(x\right)=\lim_{x\to\infty}\frac{e^{\left(\sin\frac{\pi}{10}\right)x}f\left(x\right)}{e^{\left(\sin\frac{\pi}{10}\right)x}}$

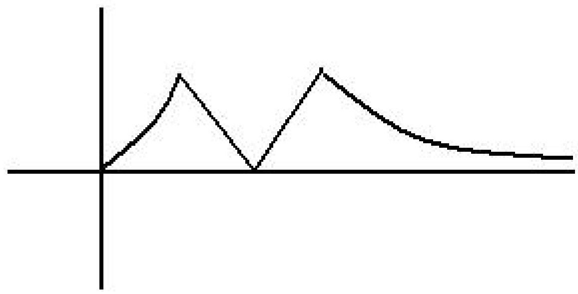
$$=\lim_{x\to\infty}\frac{e^{\left(\sin\frac{\pi}{10}\right)x}\left(f^1\left(x\right)+f\left(x\right)\sin\left(\frac{\pi}{10}\right)\right)}{e^{\left(\sin\frac{\pi}{10}\right)x}.\sin\frac{\pi}{10}}$$

$$=\frac{\sec\left(\frac{\pi}{5}\right)}{\sin\left(\frac{\pi}{10}\right)}=\frac{4}{\sqrt{5}+1}x\frac{4}{\sqrt{5}-1}=4$$

13. $\lim_{x\to\infty} f\left(x\right)=\lim_{x\to\infty}\frac{e^{\left(\tan\frac{\pi}{8}\right)\sqrt{x}}f\left(x\right)}{e^{\left(\tan\frac{\pi}{8}\right)\sqrt{x}}}$

14,15.

Setting $y=\frac{1}{x}\Rightarrow f\left(x\right)f\left(\frac{1}{x}\right)=f\left(x\right)+f\left(\frac{1}{x}\right)$ as f(1)=2. The graph of h(x) is given below. Clearly it is not diferentiable at three points.



16. (A): RHL

$$\lim_{h\to 0}\frac{\left(\frac{\pi}{2}-\sin^{-1}\left(1-h^2\right)\right)\sin^{-1}\left(1-h\right)}{\sqrt{2}\left(h-h^3\right)}$$

$$\lim_{h\to 0}\frac{\cos^{-1}\left(1-h^2\right)\sin^{-1}\left(1-h\right)}{\sqrt{2h}\left(1-\left(h\right)\right)^2}$$

$$\lim_{h\to 0}\frac{\sin^{-1}\sqrt{2h^2-h^2}\ \sin^{-1}\left(1-h\right)}{\sqrt{2h}\left(1-\left(h\right)\right)^2}$$

$$\lim_{h\to 0}\frac{\sin^{-1}\sqrt{2h^2-h^2}\ \sin^{-1}\left(1-h\right)}{\sqrt{2h}\left(1-\left(h\right)\right)^2}$$

$\Rightarrow k=\dfrac{\pi}{2}$

LHL.

$$\lim_{h\to 0} A\frac{\sin^{-1}(1-(1-h))\cos^{-1}(1-(1-h))}{\sqrt{2(1-h)}.h}$$

$$\lim_{h\to 0} A\frac{\sin^{-1}h\cos^{-1}h}{\sqrt{2(1-h)}.h}=A\frac{\pi}{2\sqrt{2}}$$

$$\Rightarrow A\frac{\pi}{2\sqrt{2}}=\frac{\pi}{2}\Rightarrow A=\sqrt{2}$$

$$\therefore \sin^2 k+\cos^2\left(\frac{A\pi}{\sqrt{2}}\right)=1+1=2$$

(B): $[2+5|n|\sin x]=2+[5|n|sinx]$

$y=5|n|\sin x$

No. of points of non diff.

$=2(5|n|-1)+1$

$=10|n|-1$

$10|n|-1=19$

$10|n|-1=20$

$|n|=2$

$n=\pm 2$

(C):

$$\text{RHD}: \lim_{h\to 0} h\frac{\left(\dfrac{\left(\frac{3}{4}\right)^{\frac{1}{h}}-\left(\frac{3}{4}\right)^{-\frac{1}{h}}}{\left(\frac{3}{4}\right)^{\frac{1}{h}}+\left(\frac{3}{4}\right)^{-\frac{1}{h}}}\right)}{h}=-1$$

$$\text{LHD}=\lim_{h\to 0}(-h)\frac{\dfrac{\left(\frac{3}{4}\right)^{-\frac{1}{h}}-\left(\frac{3}{4}\right)^{\frac{1}{h}}}{\left(\frac{3}{4}\right)^{-\frac{1}{h}}-\left(\frac{3}{4}\right)^{\frac{1}{h}}}}{-h}=1$$

$$\therefore f'(0^-)-f'(0^+)=2=P$$

Now $\displaystyle \lim_{x\to 2^-}\frac{\left(exp(x+2)\ln 4\right)^{\frac{[x+1]}{4}}-16}{4^x-16}=\frac{1}{2}$

17. $$f(x)=\begin{cases}(x+2)^{x^2+\sin x}, & 1<x<2^{\frac{1}{3}}\\ (x+2)^{x^2+\sin x}, & 2^{\frac{1}{3}}\le x<3^{\frac{1}{3}}\\ (x+4)^{x^2+\sin x}, & 3^{\frac{1}{3}}\le x<\frac{3}{2}\end{cases}$$

It is discontinuous at $x=2^{1/3}$ and $3^{1/3}$ and hence non differentiable

18. Setting $x=y=1\Rightarrow f(1)=0$

$$f^1(x)=\underset{h\to 0}{\text{Lt}}\frac{f\left(h\left(1+\frac{x}{h}\right)\right)-f(x)}{h}$$

$$=\underset{h\to 0}{\text{Lt}}\frac{e^{x\left(1+\frac{h}{x}\right)-x-1-\frac{h}{x}}\left\{e^{1+\frac{h}{x}}f(x)+e^x\left(1+\frac{h}{x}\right)\right\}-f(x)}{h}$$

$$=\underset{h\to 0}{\text{Lt}}\frac{f(x)\left(e^h-1\right)+e^{h-1-\frac{h}{x}+x}\left\{f\left(1+\frac{h}{2}\right)-f(1)\right\}}{h}$$

$$=f(x)+\frac{e^{x-1}f^1(1)}{x}$$

$$\Rightarrow f^1(x)-f(x)=\frac{e^x}{x}\Rightarrow e^{-x}\left(f^1(x)-f(x)\right)=\frac{1}{x}$$

$$\Rightarrow e^{-x}f(x)=\ln|x|+c$$

$f(1)=0\Rightarrow c=0$

$\Rightarrow f(x)=e^x\ln|x|$

$f(e)=e^e$

$\Rightarrow \ln(f(e))=e\Rightarrow[\ln(f(e))]=2$

19. We have

$$I_n = 2\int_0^1 x\left(1+\frac{x^2}{2}+\frac{x^4}{4}+\frac{x^6}{6}+...+\frac{x^{2n}}{2n}\right)dx$$

$$\left(\int_{-1}^{1}(odd)\,dx = 0\right)$$

$$= 2\left[\frac{x^2}{1.2}+\frac{x^4}{2.4}+\frac{x^6}{4.6}+.....+\frac{x^{2n+2}}{2n(2n+2)}\right]_0^1$$

$$= 2\left[\frac{1}{1.2}+\frac{1}{2.4}+\frac{1}{4.6}+.....+\frac{1}{2n(2n+2)}\right]$$

$$=1+\frac{1}{2}\left[\left(1-\frac{1}{2}\right)+\left(\frac{1}{2}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+......+\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]$$

Hence $\lim_{n\to\infty}\left[1+\frac{1}{2}\left(1-\frac{1}{n+1}\right)\right] = \frac{3}{2}$

$P = 3;\ q = 2$

SUBJECTIVE QUESTIONS

20. $f(t) = t^3 - t^2 + t + 1$

$\therefore\ f'(t) = 3t^2 - 2t + 1$

its disc $= (-2)^2 - 4.3.1 = -8 < 0$ and coefficient of $t^2 = 3 > 0$

Hence $f'(t) > 0$ for all real t. $\Rightarrow$ f(t) is always increasing

Thus f(t) is maximum when t is maximum and $t_{max} = x$

$\therefore$ max f(t) = f(x) $\Rightarrow$ g(x) =

$$\begin{cases} x^3 - x^2 + x + 1; 0 \le x \le 1 \\ 3 - x \quad ; \quad 1 < x \le 2 \end{cases}$$

Now it can be easily seen that f(x) is continuous in (0, 2) and differentiable in (0, 2) except at x = 1 because at x = 1, LHD > 0 while RHD = - 1 < 0.

21. $\frac{dz}{dx} = \frac{1}{2}\frac{\sec^2(x/2)}{\tan(x/2)} = \frac{1}{\sin x} = \operatorname{cosec} x$

$$\frac{dy}{dx} = \frac{dy}{dz}.\frac{dz}{dx} = \operatorname{cosec} x\frac{dy}{dz}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dz}\left(\frac{dy}{dx}\right)\frac{dz}{dx} = \frac{d}{dz}\left(\operatorname{cosec} x\frac{dy}{dz}\right)$$

cosec x =

$$\left(\operatorname{cosec} x\frac{d^2y}{dz^2} - \operatorname{cosec} x\cot x\frac{dx}{dz}\frac{dy}{dz}\right)\operatorname{cosec} x$$

$$= \operatorname{cosec}^2 x\frac{d^2y}{dz^2} - \operatorname{cosec} x\cot x\frac{dy}{dz}$$

Putting these values in the given differential equation, we have

$$\frac{d^2y}{dx^2} + \cot x\frac{dy}{dx} + 4y\operatorname{cosec}^2 x = 0$$

$\Leftrightarrow$ $\operatorname{cosec}^2 x$

$$\frac{d^2y}{dz^2} - \operatorname{cosec} x\cot x\frac{dy}{dz} + \cot x\operatorname{cosec} x\frac{dy}{dz} + 4y\operatorname{cosec}^2 x = 0$$

$\Leftrightarrow$

$$\operatorname{cosec}^2 x\left(\frac{d^2y}{dz^2} + 4y\right) = 0 \Leftrightarrow \frac{d^2y}{dz^2} + 4y = 0$$

22. $g(x) = f(f(x)) = \begin{cases} 1 + f(x), & 0 \le f(x) \le 2 \\ 3 - f(x), & 2 < f(x) \le 3 \end{cases}$

$$g(x) = \begin{cases} 2 + x, & 0 \le x \le 1 \\ 2 - x, & 1 < x \le 2 \\ 4 - x, & 2 < x \le 3 \end{cases}$$

Here, g(x) changes the inequality sign at x = 1 and x = 2, Thus to cheek the continuity of g(x) at x = 1 and x = 2.

At x = 1, L.H.L. $= \lim_{x\to1^-} g(x) = \lim_{x\to1^-}(2+x) = 3$

R. H. L. $= \lim_{x\to1^+} g(x) = \lim_{x\to1^+}(2-x) = 1$

$\therefore$ L.H.L. $\ne$ R.H.L. $\Rightarrow$ g(x) is discontinuous at $x = 1$

At x = 2, L.H.L. $= \lim_{x\to2^-} g(x) = \lim_{x\to2^-}(2-x) = 0$

R. H. L. $= \lim_{x\to2^+} g(x) = \lim_{x\to2^+}(4-x) = 2$

$\therefore$ L. H. L. $\ne$ R. H. L. $\Rightarrow$ g(x) is discontinuous at x = 2.

23. We have $f(x) = -1, -2 \le x \le 0 = x - 1,$
$0 < x \le 2$

And $g(x) = f(|x|) + |f(x)|$

Hence g(x) involves |x| and |x – 1| or |–1| = 1.

Threfore we should divide the given interval (–2,2) into the following intervals.

I_1	I_2	I_3
[–2, 2] = [–2, 0)	[0, 1)	[1, 2]
x = –ve	+ve	+ve

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$\lvert x\rvert = +ve$	$+ve$	$+ve$
$f(x) = -1$	$x-1$	$x-1$
$f(\lvert x\rvert) = \lvert x\rvert - 1$	$\lvert x\rvert - 1 = x-1$	
$= -x-1$	$\lvert x\rvert - 1 = x-1$	
$\lvert f(x)\rvert = \lvert -1\rvert = 1$	$\lvert x-1\rvert = -(x-1)$	
	$\lvert x-1\rvert = x-1$	

$\therefore$ Using above we get $g(x) = f|x| + |f(x)|$

$= -x - 1 + 1 = -x$ in I_1

$= x - 1 - (x-1) = 0$

in $I_2, I_3, 2f'(0) = 0$

Hence g(x) is defined as follows: $= 2(x-1)$, $1 \le x \le 2$

Clearly $g'(x) = -1, 0, 2$ respectively as g(x) is polynomial in x.

Also $Lg'(0) = -1; Rg'(0) = 0$ (not equal)

$Lg'(1) = 0; Rg'(1) = 2$ (not equal)

Hence g(x) is not differentiable both at $x = 0$ and $x = 1$.

24. $|f(x) - f(y)| \le |x-y|^3$ for all $x, y \in R \Rightarrow$

$$\frac{|f(x)-f(y)|}{|x-y|} \le |x-y|^2 \text{ for all } x, y \in R$$

$$\Rightarrow \lim_{y\to x}\left|\frac{f(x)-f(y)}{x-y}\right| \le \lim_{y\to x}|x-y|^2 \text{ for all } x$$

$$\Rightarrow \left|\lim_{y\to x}\frac{f(x)-f(y)}{x-y}\right| \le 0 \text{ for all } x$$

$\Rightarrow |f'(x)| \le 0$ for all x

$\Rightarrow f'(x) = 0$ for all x

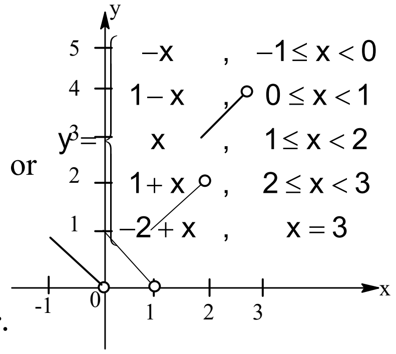
Hence f(x) is a constant function on R.

25.. We have, $= [x] + |1-x|, -1 \le x \le 3$

$$\text{or } y = \begin{cases} -1+1-x, & -1 \le x \le 0 \\ 0+1-x, & 0 \le x < 1 \\ 1-1+x, & 1 \le x < 2 \\ 2-1+x, & 2 \le x < 3 \\ 3-1+x, & x = 3 \end{cases}$$

$$\text{or } y = \begin{cases} -x, & -1 \le x < 0 \\ 1-x, & 0 \le x < 1 \\ x, & 1 \le x < 2 \\ 1+x, & 2 \le x < 3 \\ -2+x, & x = 3 \end{cases}$$

$\therefore$



The graph of the given function is as shown From graph we can say that given function is not differentiable at $x = 0, 1, 2, 3$.

$$= \lim_{h\to 0}\frac{f(x+h).f\left(\frac{2}{x}\right) - 2f(1)}{h.f\left(\frac{2}{x}\right)} = \lim_{h\to 0}\frac{2f\left(\frac{(x+h).2}{2x}\right) - 2f(1)}{h.f\left(\frac{2}{x}\right)}$$

$$= \frac{2}{f\left(\frac{2}{x}\right)}\lim_{h\to 0}\frac{f\left(1+\frac{h}{x}\right) - f(1)}{\frac{h}{x}.x} = \frac{2}{f\left(\frac{2}{x}\right)}.\frac{f'(1)}{x}$$

$$= \frac{f(x)}{f(1)}.\frac{f'(1)}{x} \Rightarrow f'(x) = \frac{f(x)}{x} \{\text{as } f'(1) = f(1)\}$$

$$\Rightarrow xf'(x) - f(x) = 0 \Rightarrow \frac{f(x) - xf'(x)}{f^2(x)} = 0$$

$$\Rightarrow \frac{d}{dx}\left(\frac{x}{f(x)}\right) = 0$$

$$\left\{\text{as; } \frac{d}{dx}\left(\frac{x}{f(x)}\right) = \frac{f(x).1 - x.f'(x)}{f^2(x)}\right\}$$

$$\Rightarrow \frac{x}{f(x)} = \lambda \text{ (constant)} \Rightarrow \frac{f(x)}{x} = \frac{1}{\lambda} \text{ or}$$

$$f(x) = \frac{x}{\lambda} \text{ and } f(1-x) = \frac{1-x}{\lambda}$$

adding the above two we get;

$$f(x) + f(1-x) = \frac{x}{\lambda} + \frac{1-x}{\lambda} = \frac{1}{\lambda}$$

$\therefore f(x) + f(1-x) = \dfrac{1}{\lambda} \Rightarrow$ which is clearly constant.

(ii)

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f(x)\left[\frac{f(x+h)}{f(x)}-1\right]}{h}$$

$$\Rightarrow f'(x)=\lim_{h\to 0}\frac{f(x)\left[f\left(1+\frac{h}{x}\right)-1\right]}{h}$$

$$\Rightarrow f'(x)=\lim_{h\to 0}f(x)\frac{\left[f\left(1+\frac{h}{x}\right)-f(1)\right]}{x\times\frac{h}{x}}$$

$(\because f(1)=1)$

$$\Rightarrow f'(x)=\frac{f(x)}{x}\cdot f'(1)\Rightarrow f'(x)=\frac{3f(x)}{x}$$

$$\Rightarrow \log(f(x))=3\log x+c$$

at $x=1$, $c=0 \Rightarrow f(x)=x^3$

26. $f(xy)=f(x)\,f(y)$ for all $x, y\in R$

$\Rightarrow f(1)=f(1)\,f(1)=f^2(1)$

$\Rightarrow f(1)=0$ or $f(1)=1$

$\Rightarrow f(1)\,\{f(1)-1\}=0$

If $f(1)=0$, then $f(x)=f(x.1)=f(x).\,f(1)$

$=f(x)\times 0=0$ for all $x\in R$

Which is every where continuous and if $f(1)=1$

It is given that f is continuous at $x=1$, then

$$\lim_{x\to 1^+}f(x)=f(1)=\lim_{x\to 1^-}f(x)$$

$$\Rightarrow \lim_{h\to 0}f(1+h)=f(1)=\lim_{h\to 0}f(1-h)$$

$$\Rightarrow \lim_{h\to 0}f(1+h)=\lim_{h\to 0}f(1-h)=1$$

Now, let a be a any nonzero real number, then

$$\lim_{x\to a^+}f(x)=\lim_{h\to 0}f(a+h)=\lim_{h\to 0}f\left\{a\left(1+\frac{h}{a}\right)\right\}$$

$$=\lim_{h\to 0}f(a)\,f\left(1+\frac{h}{a}\right)\quad [\because f(xy)=f(x)f(y)]$$

$$=f(a)\lim_{h\to 0}f\left(1+\frac{h}{a}\right)=f(a).1=f(a)$$

and $\lim_{x\to a^-}f(x)=\lim_{h\to 0}f(a-h)$

$$=\lim_{h\to 0}f\{a(1-\frac{h}{a})\}$$

$$=\lim_{h\to 0}f(a).f\left(1-\frac{h}{a}\right)\quad [\because f(xy)=f(x)f(y)]$$

$$=f(a)\lim_{h\to 0}f\left(1-\frac{h}{a}\right)=f(a).1=f(a)$$

$$\Rightarrow \lim_{x\to a^+}f(x)=f(a)=\lim_{x\to a^-}f(x)$$

$\Rightarrow f(x)$ is continuous at $x=a$. Since a is an arbitrary non–zero real number

$\Rightarrow f(x)$ is continuous for all nonzero $x\in R$.

27.. Given $f\left(\frac{x+y}{3}\right)=\frac{2+f(x)+f(y)}{3}$ ... (1)

$$\because\quad f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f\left(\frac{3x+3h}{3}\right)-f\left(\frac{3x+0}{3}\right)}{h}$$

$$f'(x)=\lim_{h\to 0}\left(\frac{\frac{2+f(3x)+f(3h)}{3}-\frac{2+f(3x)+f(0)}{3}}{h}\right)$$

using (1)

$$f'(x)=\lim_{h\to 0}\frac{f(3h)-f(0)}{3h}=\lim_{h\to 0}\frac{f(0+3h)-f(0)}{3h}=f'(0)$$

$\therefore\quad f'(x)=f'(0)=2\Rightarrow$

$$\int f'(x).dx=2\int dx$$

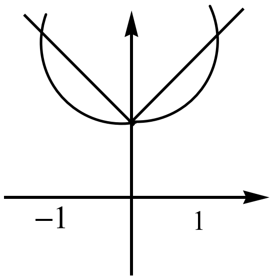
$[\because f\left(\frac{x+y}{3}\right)=\frac{2+f(x)+f(y)}{3}$ putting $x=y=0$

$$\Rightarrow f(0)=\frac{0+2f(0)}{3}\Rightarrow f(0)=2]$$

$\Rightarrow f(x)=2x+c$, Putting $f(0)=c=2$

$\therefore f(x)=2x+2$

28.
$$h(x)=\begin{cases}x^2+1, & x\in(-\infty,-1]\\ -x+1, & x\in[-1,0]\\ x^2+1, & x\in[0,1]\\ x+1, & x\in[1,\infty)\end{cases}$$



$h(x)$ is not differentiable at $x = -1, 0, 1$

29. We have, $x_n^2 = a + x_{n-1}$.
It is easy to see that the variable x^n increases. Let us show that all its values remain less than some constant number. We have

$x_{n-1}^2 - x_{n-1} - a < 0 \qquad (\because x_{n-1} < x_n)$

Hence,

$$\left(x_{n-1} - \frac{\sqrt{(4a+1)}+1}{2}\right)\left(x_{n-1} + \frac{\sqrt{(4a+1)}-1}{2}\right) < 0$$

But since the second bracketed expression exceeds by zero,

it must be $x_{n-1} < \dfrac{\sqrt{4(a+1)}+1}{2}$

Put $\lim\limits_{n\to\infty} x_{n-1} = \lim\limits_{n\to\infty} x_n = \alpha$

From the origin relation between x_n and x_{n-1}, we get $\alpha^2 - \alpha - a = 0$

$$\therefore \ \alpha = \frac{1 \pm \sqrt{(1+4a)}}{2}$$

and since $\alpha \geq 0$, we have $\alpha = \dfrac{\sqrt{(1+4a)}+1}{2}$

30. $f(x+y) = f(x)\cdot f(y)$

and $f'(0) = \lim\limits_{h\to 0} \dfrac{f(h)-f(0)}{h} = 2$

$$\Rightarrow \ f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f(x+h)-f(x+0)}{h}$$

$$\Rightarrow \ f'(x) = \lim_{h\to 0}\frac{f(x)\cdot f(h) - f(x)\cdot f(0)}{h}$$

$$= f(x)\lim_{h\to 0}\frac{f(h)-f(0)}{h}$$

$\Rightarrow \ f'(x) = 2f(x) \ \Rightarrow \ \ln f(x) = 2x + c$

$\Rightarrow \ f(x) = e^{2x} + c$ at $x = 0$, $c = 0$

$(\because f(0) = 1) \ \Rightarrow \ f(x) = e^{2x}$

31. $f(x) = \sqrt{|x-1|}$

$g(x) = \sin x$

$\therefore \ f(g(x)) = \sqrt{|\sin x - 1|}$

$g(f(x)) = \sin\sqrt{|x-1|} = h(x)$ (say)

$$h'(x) = \cos\sqrt{|x-1|} \times \frac{1}{2\sqrt{|x-1|}}$$

$\therefore$ $h'(x)$ does not exist at $x = 1$

32. $f\left(\dfrac{x+y}{2}\right) = \dfrac{f(x)+f(y)}{2}$ for all $x, y \in R$

$f\left(\dfrac{x}{2}\right) = \dfrac{f(x)+f(0)}{2}$ for all $x \in R$

(putting $y = 0$)

$f\left(\dfrac{x}{2}\right) = \dfrac{f(x)+1}{2}$ for all $x \in R$ $\ (\because f(0) = 1)$

$\Rightarrow f(x) = 2f\left(\dfrac{x}{2}\right) - 1$ for all $x \in R$ $\quad \ldots (1)$

Now, $f'(x) = \lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$f'(x) =$

$$\lim_{h\to 0}\frac{\left\{2f\left(\frac{x+h}{2}\right)-1\right\} - \left\{2f\left(\frac{x}{2}\right)-1\right\}}{h}$$

$$f'(x) = 2\lim_{h\to 0}\frac{f\left(\frac{x+h}{2}\right) - f\left(\frac{x}{2}\right)}{h}$$

$$f'(x) = \lim_{h\to 0}\frac{\frac{f(x)+f(h)}{2} - f\left(\frac{x}{2}\right)}{h}$$

$$\left[\because f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}\right]$$

$$f'(x) = \lim_{h\to 0}\frac{f(x)+f(h)-2f\left(\frac{x}{2}\right)}{h}$$

$$f'(x)=\lim_{h\to 0}\frac{f(h)-1}{h} \quad \text{(by (1))}$$

$$f'(x)=\lim_{h\to 0}\frac{f(h)-f(0)}{h} \quad [\because f(0)=1]$$

$$f'(x)=f'(0) \qquad [\because f'(0)=-1]$$

$\Rightarrow f'(x)=-1$

$\Rightarrow f(x)=-x+c$ for all $x\in R \Rightarrow f(0)=c$

$\Rightarrow c=1$

$\therefore f(x)=-x+1 \Rightarrow f(2)=-2+1=-1$

33. i) $\lim_{x\to 0}\frac{2(\tan x-\sin x)-x^3}{x^5}$

$$=\lim_{x\to 0}\frac{2\left(x+\frac{x^3}{3}+\frac{2}{15}x^5+\ldots-x+\frac{x^3}{\lfloor 3}-\frac{x^5}{\lfloor 5}\right)-x^3}{x^5}$$

$$=\lim_{x\to 0}\frac{\frac{2x^3}{2}+\frac{30}{120}x^5-x^3}{x^5}=\frac{30}{120}=\frac{1}{4}$$

ii) $\lim_{x\to 1}\frac{(\ln(1+x)-\ln 2)(3.4^{x-1}-3x)}{[(7+x)^{1/3}-(1+3x)^{1/2}].\sin(x-1)}$

$$=\lim_{x\to 1}\frac{\ln\left(\frac{1+x}{2}\right).\left(4^{x-1}-x\right)\times\left((7+x)^{1/3}+(1+3x)^{1/2}\right)}{\sin(x-1)\times((7+x)^{2/3}-1-3x)}$$

$$=3\lim_{x\to 1}\frac{\ln\left(1+\frac{x-1}{2}\right)(4^{x-1}-x)\times(2+2)}{\sin(x-1)((7+x)^{2/3}-(1+3x))}$$

$$=12\lim_{x\to 1}\frac{\ln\left(1+\frac{x-1}{2}\right)}{\sin(x-1)}\times\frac{(4^{x-1}-1-(x-1))}{(7+x)^{2/3}-8^{2/3}+3-3x}$$

$x-1=t \Rightarrow x\to 1, t\to 0$

$$=12\lim_{t\to 0}\frac{\ln\left(1+\frac{t}{2}\right)}{\sin t}\times\frac{4^t-1-t}{(t+8)^{2/3}-8^{2/3}-t}$$

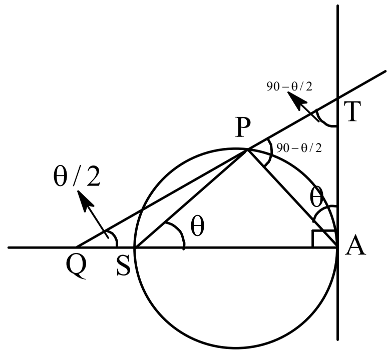
$$=12\lim_{t\to 0}\frac{\ln\left(1+\frac{t}{2}\right)}{2\times\frac{t}{2}}\times\frac{t}{\sin t}\times\frac{\frac{4^t-1}{t}-1}{(t+8)^{2/3}-8^{2/3}-t}$$

$$=12\times\frac{\frac{1}{2}}{1}\times\frac{\ln 4-1}{\frac{2}{3}.(8)^{\frac{2}{3}-1}-1} = 12\times\frac{1}{2}\times\frac{\ln(4/e)}{\frac{2}{3}\times\frac{1}{2}-1}$$

$$=12\times\frac{1}{2}\times\frac{\ln 4/e\times 3}{\frac{-4}{3}}$$

$=-9\ln(4/e)$

iii)



$$\tan\left(\frac{\theta}{2}\right)=\frac{AT}{AQ} \Rightarrow AT=AQ\tan\frac{\theta}{2} \qquad \ldots(1)$$

$$\sin\theta=\frac{AP}{2r}$$

$\Rightarrow AP=D\sin\theta$ (where D is the diameter of circle)… (2)

$\Rightarrow AQ\tan\theta/2=2D\sin\theta/2\cos\theta/2$

$\Rightarrow AQ=2D\cos^2\theta/2$

$\lim_{\theta\to 0}AQ=2D\lim_{\theta\to 0}\cos^2\theta/2 \Rightarrow AQ=2D$

$=2\times$ Diameter of circle

34. $g(x)$ is continuous at $x=\alpha$

$\Rightarrow \lim_{x\to\alpha}g(x)=g(\alpha)$ and $f(x)-f(\alpha)=g(x)$ $(x-\alpha)$, $x\in R$.

$$\Rightarrow \frac{f(x)-f(\alpha)}{(x-\alpha)}=g(x)$$

$$\Rightarrow \lim_{x\to\alpha}\frac{f(x)-f(\alpha)}{(x-\alpha)}=\lim_{x\to\alpha}g(x) \Rightarrow f'(\alpha)=g(\alpha)$$

Therefore, $f(x)$ is differentiable at $x=\alpha$

Therefore, $f(x)$ is differentiable at $x=\alpha$

Conversely, f is differentiable at $x=\alpha$, then

$\lim_{x\to\alpha}\frac{f(x)-f(\alpha)}{(x-\alpha)}$ exists finitely.

Let $g(x)=\begin{cases}\dfrac{f(x)-f(\alpha)}{x-\alpha}, & x\neq\alpha\\ f'(\alpha), & x=\alpha\end{cases}$

Clearly $\lim\limits_{x\to\alpha} g(x)=f'(x)$

Hence, $g(x)$ is continuous at $x=\alpha$.
Therefore $f(x)$ is differentiable at $x=\alpha$
Iff $g(x)$ is continuous at $x=\alpha$.

35. Given that f: $[-2a, 2a]\to R$
f is an odd function.

Lf′ at x = a is 0. $\Rightarrow \lim\limits_{h\to 0}\dfrac{f(a-h)-f(a)}{-h}=0$

$\Rightarrow \lim\limits_{h\to 0}\dfrac{f(a-h)-f(a)}{h}=0 \qquad \ldots (1)$

To find Lf′ at x = –a which is given by

$\lim\limits_{h\to 0}\dfrac{f(-a-h)-f(a)}{-h}=\lim\limits_{h\to 0}\dfrac{-f(a+h)+f(a)}{-h}$

$[\because f(-x)=-f(x)]$

$=\lim\limits_{h\to 0}\dfrac{-f(a+h)+f(a)}{-h}=\lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$

Again for $x\in[a, 2a]$
$f(x)=f(2a-x)$
$\therefore f(a+h)=f(2a-a-h)=f(a-h)$
Substititing this values in last expression we get

$Lf'(-a)=\lim\limits_{h\to 0}\dfrac{f(a-h)-f(a)}{h}=0$

[Using equation (1)]
Hence $Lf'(-a)=0$

36. To find, $\lim\limits_{n\to\infty}\left[(n+1)\dfrac{2}{\pi}\cos^{-1}\left(\dfrac{1}{n}\right)-n\right]$

$=\lim\limits_{n\to\infty} n\left[\left(1+\dfrac{1}{n}\right)\dfrac{2}{\pi}\cos^{-1}\left(\dfrac{1}{n}\right)-1\right]=\lim\limits_{n\to\infty} nf\left(\dfrac{1}{n}\right)$

where $f(x)=\left[(1+x)\dfrac{2}{\pi}\cos^{-1}x-1\right]$

$f(0)=\left[(1+0)\dfrac{2}{\pi}\cos^{-1}0-1\right]=\dfrac{2}{\pi}\cdot\dfrac{\pi}{2}-1=0$

$\therefore$ Using given relation as $\lim\limits_{n\to\infty} nf\left(\dfrac{1}{n}\right)=f'(0)$

then given limit becomes

$=f'(0)=\dfrac{d}{dx}\left[(1+x)\dfrac{2}{\pi}\cos^{-1}x-1\right]_{x=0}$

$=\dfrac{2}{\pi}\left[\cos^{-1}x-\dfrac{1+x}{\sqrt{1-x^2}}\right]_{x=0}$

$=\dfrac{2}{\pi}\left[\dfrac{\pi}{2}-1\right]=1-\dfrac{2}{\pi}=\dfrac{\pi-2}{\pi}$.

37. Given that,
$f(x-y)=f(x).g(y)-f(y).g(x) \quad \ldots (i)$
$g(x-y)=g(x).g(y)+f(x)f(y) \quad \ldots (ii)$
In equ. (i) putting x = y we get
$f(0)=f(x)g(x)-f(x)g(x) \Rightarrow f(0)=0$
Putting y = 0 in eqn. (i), we get
$f(x)=f(x)g(0)-f(0)g(x)\Rightarrow f(x)=f(x)g(0)$
[using $f(0)=0$] $\Rightarrow g(0)=1$
putting x = y in eqn. (ii), we get
$g(0)=g(x)g(x)+f(x)f(x)$
$\Rightarrow 1=[g(x)]^2+[f(x)]^2$ [using $g(0)=1$]
$\Rightarrow g(0)=1$
putting x = y in eqn. (ii), we get $g(0)=g(x)$
$g(x)+f(x)f(x)\Rightarrow 1=[g(x)]^2+[f(x)]^2$
[using $g(0)=1$]
$\Rightarrow [g(x)]^2=1-[f(x)]^2\ldots$ (iii)
clearly g(x) will be differentiable only if f(x) is differentiable.
$\therefore$ first we will check the differentiability of f(x)

Given that Rf′(0) exists i.e., $\lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}$
exists

i.e., $\lim\limits_{h\to 0}\dfrac{f(0)g(-h)-f(-h)g(0)}{h}$ exists i.e.,

$\lim\limits_{h\to 0}\dfrac{-f(-h)}{h}$ exists

(using $f(0)=0$ and $g(0)=1$
which can be written as, $\Rightarrow$

$\lim\limits_{h\to 0}\dfrac{f(0)-f(-h)}{-h}=Lf'(0)\Rightarrow Lf'(0)=Rf'(0)$

$\therefore$ f is differentiable, at x = 0
Differentiating equation (iii) we get
$2g(x).g'(x)=-2f(x).f'(x)$

for x = 0

$\Rightarrow g(0) \cdot g'(0) = -f(0)f'(0) \Rightarrow g'(0) = 0$

[using f(0) = 0 and g(0) = 1].

38. First check the differentiabilty:

$$Lf'(0) = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \lim_{h\to 0}\frac{(0-h)\left\{1+\frac{1}{3}\sin\left(\ln h^2\right)\right\}-0}{(-h)}$$

$$= \lim_{h\to 0}\left\{1+\frac{1}{3}\sin\left(\ln h^2\right)\right\}$$

$$= 1+\frac{1}{3}\sin(\infty) = 1+\frac{1}{3}(lies\ between -1 to 1)$$

= Does not exist.

(Similarly $Rf'(0)$ also does not exist)

Hence $f(x)$ is not differentiable at $x=0$.

Check the continuity at $x=0$:

LHL= $\lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h)$

$$= \lim_{h\to 0} f(0-h)\left\{1+\frac{1}{3}\sin\left(\ln h^2\right)\right\}$$

$$= 0\left\{1+\frac{1}{3}\sin\infty\right\} = 0$$

RHL= $\lim_{x\to 0^+} f(x)$

$$= \lim_{x\to 0^+} f(0+h)$$

$$= \lim_{h\to 0} h\left\{1+\frac{1}{3}\sin\left(\ln h^2\right)\right\}$$

$$= 0\left\{0+\frac{1}{3}\sin\infty\right\} = 0$$

LHL=RHL=0

Finally $f(x)$ is continuous at $x=0$ but not differentiable at $x=0$.

39. RHL $= \lim_{x\to x^+} f(x) = \lim_{h\to 0} f(x+h)$

$$= \lim_{h\to 0} f(x)f(h) - \sqrt{6-f(h)}$$

$$= f(x)\lim_{h\to 0} f(h) - \lim_{h\to 0}\sqrt{6-f(h)}$$

$$= f(x).6 - 0 = 6f(x)$$

$$\neq f(x)$$

$\neq$ VF

This shows that if $f(x)\neq 0$, then f is discontinuous at x. If $f(x)=0$, then $f(x)$ is continuous at x.

40. Given $f(x+f(y)) = f(f(x)) + f(y)$

putting $x=y=0$ in(i), then

$$f(0+f(0)) = f(f(0)+f(0))$$

$$\Rightarrow f(f(0)) = f(f(0)) + f(0)$$

$$\therefore f(0) = 0$$

Now $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$$= \lim_{h\to 0}\frac{f(h+x)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{f(f(h))}{h}$$

$$= \lim_{h\to 0}\frac{f(h)}{h}$$

$$= \lim_{h\to 0}\frac{h}{h} = 1$$

Integrating both sides with limits 0 to x and then $f(x) = x$ $\quad \therefore f'(x) = 1$

41. Let given limit = L, then

$$L = \lim_{n\to\infty}\left(\frac{1}{2n+1}+\frac{1}{2n+2}+\frac{1}{2n+3}+.....+\frac{1}{4n}\right) -$$

$$\lim_{n\to\infty}\left(\frac{1}{2n+2}+\frac{1}{2n+4}+\frac{1}{2n+6}+.....+\frac{1}{4n}\right)$$

$$= \lim_{n\to\infty}\left[\frac{1}{n}\sum_{r=1}^{2n}\frac{n}{2n+r} - \frac{1}{n}\sum_{r=1}^{n}\frac{n}{2n+2r}\right]$$

$$= \lim_{n\to\infty}\left[\frac{1}{n}\sum_{r=1}^{2n}\frac{n}{2+\frac{r}{n}} - \frac{1}{n}\sum_{r=1}^{n}\frac{n}{2+2\left(\frac{r}{n}\right)}\right]$$

$$= \int_0^2 \frac{1}{2+x}\,dx - \int_0^1 \frac{1}{2+2x}\,dx$$

$$= \left[\ln(2+x)\right]_0^2 - \frac{1}{2}\left[\ln(1+x)\right]_0^1$$

$$= \ln 4 - \ln 2 - \frac{1}{2}\ln 2 = \left(2-\frac{3}{2}\right)\ln 2$$

$$= \frac{1}{2}\ln 2 = \frac{a}{b}\ln c$$

$\therefore$ Least sum $a+b+c = 1+2+2 = 5$.